Biostatistics-HW

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1

(a)

```
library(irr)
```

```
Warning: package 'irr' was built under R version 4.3.2

Loading required package: lpSolve

Warning: package 'lpSolve' was built under R version 4.3.2
```

```
# The actual data from the table provided
biopsy_data <- matrix(c(83, 37, 2, 1, 0,
                        18, 79, 8, 9, 0,
                        0, 10, 4, 10, 1,
                        0, 0, 2, 6, 2,
                        0, 0, 0, 1, 2),
                      byrow = TRUE, nrow = 5)

biopsy_grades <- expand.grid(Reader1 = 6:10, Reader2 = 6:10)
biopsy_grades$Freq <- c(biopsy_data)

biopsy_ratings <- biopsy_grades[rep(1:nrow(biopsy_grades), biopsy_grades$Freq), 1:2]

# Calculate unweighted Kappa
kappa_unweighted <- kappa2(biopsy_ratings, weight = "unweighted")
print(kappa_unweighted)
```

```
Cohen's Kappa for 2 Raters (Weights: unweighted)

Subjects = 275
  Raters = 2
   Kappa = 0.423

       z = 10.1
 p-value = 0
```

```
# Calculate quadratically weighted Kappa
kappa_weighted <- kappa2(biopsy_ratings, weight = "squared")
```

```
print(kappa_weighted)
```

```
Cohen's Kappa for 2 Raters (Weights: squared)

Subjects = 275
  Raters = 2
   Kappa = 0.536

       z = 9.14
 p-value = 0
```

(b)

H_0 There is no agreement beyond chance k_w = 0

H_a There is agreement beyond chance $k_w \neq$ 0

The p-value for both the unweighted and the quadratically weighted Kappa is reported as 0. This indicates that the agreement between the two readers is statistically significantly different from what would be expected by chance alone, at any conventional significance level. The hypothesis tests for both the unweighted and the quadratically weighted Kappa reject the null hypothesis of no agreement beyond chance. Thus, we can conclude that there is a statistically significant agreement between the two raters' grading of prostate cancer biopsies.

2

(a)

```
# Original 5x5 table of grades
original_table <- matrix(c(83, 18, 0, 0, 0,
                           37, 79, 10, 0, 0,
                           2, 8, 4, 2, 0,
                           1, 9, 10, 6, 1,
                           0, 0, 1, 2, 2), byrow = TRUE, nrow = 5)

# Reclassify into binary categories (<9 vs. ≥9)
a <- sum(original_table[1:3, 1:3])  # True Negatives (TN)
b <- sum(original_table[1:3, 4:5])  # False Positives (FP)
c <- sum(original_table[4:5, 1:3])  # False Negatives (FN)
d <- sum(original_table[4:5, 4:5])  # True Positives (TP)

# Construct the 2x2 table
table_2x2 <- matrix(c(a, b, c, d), byrow = TRUE, nrow = 2)
colnames(table_2x2) <- c("<9", "≥9")
```

```
rownames(table_2x2) <- c("<9", "≥9")
table_2x2
```

```
    <9 ≥9
<9 241  2
≥9  21 11
```

(b)

```
# Sensitivity and Specificity
sensitivity <- d / (c + d)
specificity <- a / (a + b)
sensitivity
```

```
[1] 0.34375
```

```
specificity
```

```
[1] 0.9917695
```

- This indicates that Reader 2 correctly identified 34.38% of the high-risk tumors as classified by the gold standard.
- This indicates that Reader 2 correctly identified 99.18% of the non-high-risk tumors.

Reader 2 is highly specific in identifying non-high-risk tumors but not as sensitive in identifying high-risk tumors. This means that when Reader 2 identifies a tumor as non-high-risk, they are likely correct. However, Reader 2 misses a significant number of high-risk tumors.

(c)

```
# Accuracy
accuracy <- (a + d) / sum(table_2x2)
accuracy
```

```
[1] 0.9163636
```

```
# Odds Ratio (OR)
odds_ratio <- (d / b) / (c / a)
odds_ratio
```

```
[1] 63.11905
```

```
# Youden's Index
youens_index <- sensitivity + specificity - 1
```

```
youens_index
```

```
[1] 0.3355195
```

- Accuracy: Approximately 91.64%. This suggests that overall, Reader 2 correctly identified whether a tumor was high-risk or not in 91.64% of the cases.
- Odds Ratio (OR): Approximately 63.12. This indicates that Reader 2 is about 63 times more likely to correctly identify a high-risk tumor compared to incorrectly identifying a non-high-risk tumor, relative to chance.
- Youden's Index (YI): Approximately 0.34. This index indicates that Reader 2's ability to identify high-risk tumors is above chance but not perfect. The closer this index is to 1, the better the diagnostic ability.

The Youden's Index suggests that Reader 2 has a moderate ability to distinguish between high-risk and non-high-risk tumors.

Although there are fewer high-risk cases, the high specificity mostly drives the high accuracy, which is still encouraging.

3 (a)

(i)

MODEL 1

$log(\pi_i/(1-\pi_i))$ = $\beta_0 + \beta_1 X_i$

MODEL 2

$log(\pi_i/(1-\pi_i))$ = $\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3}$

(ii)

The null hypothesis for the LRT is that Model 1 (simpler model) fits the data as well as Model 2 (more complex model).

The alternative hypothesis is that Model 2 fits the data significantly better than Model 1.

```
library(stats)

# Residual deviance from the outputs
residual_deviance_model_1 <- 3675.3
residual_deviance_model_2 <- 3673.3

# Degrees of freedom for Model 1 and Model 2
# Model 1 has 2 parameters (intercept and BMI), Model 2 has 4 parameters (intercept and 3 dummy va
df_model_1 <- 46
df_model_2 <- 43
```

```r
# Calculate the test statistic D
D <- residual_deviance_model_1 - residual_deviance_model_2

# Degrees of freedom for the test
df <- df_model_1 - df_model_2

# Calculate the p-value
p_value <- pchisq(D, df, lower.tail = FALSE)

c(D, p_value)
```

```
[1] 2.0000000 0.5724067
```

The test statistic is: D=3675.3−3673.3=2.0

The degrees of freedom for the test are: df=45−43=2

The likelihood ratio test statistic is 2.0, and the corresponding p-value is approximately 0.368. Since this p-value is greater than the conventional alpha level of 0.05, we do not reject the null hypothesis.

This means that we do not have sufficient evidence to conclude that Model 2 provides a significantly better fit to the data than Model 1. Therefore, for simplicity, we would typically choose Model 1 for further interpretation.

In Model 1, the estimated regression coefficient for BMI is β_1 = 0.20073. This implies that for each unit increase in BMI, the log-odds of colorectal cancer increases by 0.20073, holding all else constant. To interpret this in terms of odds ratio, we would exponentiate the coefficient which indicates the multiplicative change in the odds of colorectal cancer for a one-unit increase in BMI.

(b)

```r
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.3.2

Warning: package 'ggplot2' was built under R version 4.3.3

Warning: package 'tidyr' was built under R version 4.3.3

Warning: package 'dplyr' was built under R version 4.3.3

── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.1.4     ✓ readr     2.1.4
✓ forcats   1.0.0     ✓ stringr   1.5.0
✓ ggplot2   3.5.0     ✓ tibble    3.2.1
✓ lubridate 1.9.2     ✓ tidyr     1.3.1
✓ purrr     1.0.2
```

```
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
errors
```

```r
# Coefficient for BMI from Model 1
coef_BMI_model_1 <- 0.20073

# Calculate the odds ratio
odds_ratio_BMI_model_1 <- exp(coef_BMI_model_1)

odds_ratio_BMI_model_1
```

```
[1] 1.222295
```

The odds ratio for BMI from Model 1 is approximately 1.222. This means that for each one-unit increase in BMI, the odds of colorectal cancer increase by about 22.2%, assuming other factors are held constant.

Model 3 (BMI and Age): BMI coefficient = 0.21621

Model 4 (BMI and Smoking Status): BMI coefficient = 0.1837

Model 5 (BMI, Age, and Smoking Status): BMI coefficient = 0.20381

```r
# Coefficients for BMI from the different models
coef_model1 <- 0.20073
coef_model3 <- 0.21621
coef_model4 <- 0.1837
coef_model5 <- 0.20361

# Calculating the odds ratios for BMI from each model's coefficient
or_model1 <- exp(coef_model1)
or_model3 <- exp(coef_model3)
or_model4 <- exp(coef_model4)
or_model5 <- exp(coef_model5)

# The percentage change in odds ratio when adjusting for other variables
# Comparing with the crude estimate from Model 1
percent_change_model3 <- ((or_model3 - or_model1) / or_model1) * 100
percent_change_model4 <- ((or_model4 - or_model1) / or_model1) * 100
percent_change_model5 <- ((or_model5 - or_model1) / or_model1) * 100

list(c(or_model1, or_model3, percent_change_model3),
     c(or_model1, or_model4, percent_change_model4),
     c(or_model1, or_model5, percent_change_model5))
```

```
[[1]]
[1] 1.222295 1.241363 1.560044
```

```
[[2]]
[1]  1.222295  1.201655 -1.688581

[[3]]
[1] 1.2222947 1.2258200 0.2884151
```

- Model 1 (crude estimate): OR for BMI = 1.2223 Model 3 (adjusted for Age): OR for BMI = 1.2414, which is approximately 1.56% higher than the crude estimate.
- Model 4 (adjusted for Smoking status): OR for BMI = 1.2017, which is approximately 1.69% lower than the crude estimate.
- Model 5 (adjusted for Age and Smoking status): OR for BMI = 1.2258, which is approximately 0.29% higher than the crude estimate.

Based on these results:

The adjustment for age alone (Model 3) resulted in a slight increase in the estimated OR for BMI, suggesting a small degree of confounding by age, but this change is relatively minor (around 1.56%).

The adjustment for smoking status alone (Model 4) resulted in a slight decrease in the estimated OR for BMI, suggesting a small degree of confounding by smoking status, but this change is also relatively minor (around -1.69%).

The adjustment for both age and smoking status (Model 5) resulted in a negligible change in the estimated OR for BMI (around 0.29% increase), suggesting that there is no meaningful confounding by the collection of these variables when controlled simultaneously.

In conclusion, the individual adjustments for age and smoking status cause only a small change in the estimated OR for BMI, and the simultaneous adjustment for both variables in Model 5 does not result in a meaningful change in the estimated magnitude of association. Thus, we may not consider age and smoking status as significant confounders in the association between BMI and the risk of colorectal cancer, according to the given data and the usual 10% change threshold for confounding.

(c)

$log(\pi_i/(1-\pi_i))$ = $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$

$H_0 : \beta_1$ =0 (BMI is not associated with the risk of colorectal cancer after controlling for age and smoking status).

$H_a : \beta_1 \neq 0$ (BMI is associated with the risk of colorectal cancer after controlling for age and smoking status).

$Z = \beta_1 / SE(\beta_1)$

```
coef_bmi <- 0.20361
std_err_bmi <- 0.07085
```

```
# Calculating the Wald test statistic (z value)
z_value <- coef_bmi / std_err_bmi

# Calculating the p-value from the Wald test statistic
p_value_bmi <- 2 * (1 - pchisq(z_value^2, 1))  # Two-tailed test

cat("z_value:", z_value, "\np_value_bmi:", p_value_bmi)
```

```
z_value: 2.873818
p_value_bmi: 0.008110854
```

The Wald test statistic for the BMI coefficient is approximately 2.874 with a corresponding p-value of about 0.0081. Since the p-value is less than the conventional significance level of 0.05, we reject the null hypothesis.

In conclusion, there is statistically significant evidence to suggest that BMI is associated with the risk of colorectal cancer after controlling for age and smoking status in Model 5.

(d)

(i)

The odds ratio (OR) for a one-unit increase in BMI can be calculated by exponentiating the coefficient of BMI

```
# Coefficients from Model 5
beta_BMI <- 0.20361
SE_BMI <- 0.07085

# Calculating the odds ratio for a one-unit increase in BMI
OR <- exp(beta_BMI)

# Calculating the 95% confidence interval for the OR
lower_CI <- exp(beta_BMI - 1.96 * SE_BMI)
upper_CI <- exp(beta_BMI + 1.96 * SE_BMI)

# Since we are interested in the change from healthy-weight to overweight, which involves a chang
# we adjust the OR for this specific change.
adjusted_OR <- OR^4

# Adjusted confidence interval for the change
adjusted_lower_CI <- lower_CI^4
adjusted_upper_CI <- upper_CI^4

# Printing the results
cat("Odds Ratio for a one-unit increase in BMI:", OR, "\n",
    "95% Confidence Interval for OR:", lower_CI, "-", upper_CI, "\n",
```

```
    "Adjusted Odds Ratio for moving from healthy-weight to overweight:", adjusted_OR, "\n",
    "Adjusted 95% Confidence Interval:", adjusted_lower_CI, "-", adjusted_upper_CI, "\n")
```

```
Odds Ratio for a one-unit increase in BMI: 1.22582
 95% Confidence Interval for OR: 1.066886 - 1.408431
 Adjusted Odds Ratio for moving from healthy-weight to overweight: 2.257911
 Adjusted 95% Confidence Interval: 1.295603 - 3.934973
```

For the overweight range (25-29.99 kg/m^2), we can consider the increase from the midpoint of the healthy-weight range (23.495 kg/m^2) to the midpoint of the overweight range (27.495 kg/m^2), which is an increase of 4 BMI units.

The odds ratio (OR) for a one-unit increase in BMI is approximately 1.23, with a 95% confidence interval (CI) ranging from 1.07 to 1.41. This means that, holding age and smoking status constant, each one-unit increase in BMI is associated with a 23% increase in the odds of colorectal cancer, with a confidence interval suggesting that the true increase in odds could be between 7% and 41%.

This wide interval reflects the uncertainty around the exact magnitude of the increased odds, but consistently suggests a significant increase in the odds of colorectal cancer for overweight males relative to their healthy-weight counterparts.

(ii) Obese

```r
# Coefficients from Model 5
beta_BMI <- 0.20361
SE_BMI <- 0.07085

# Calculating the odds ratio for a one-unit increase in BMI
OR <- exp(beta_BMI)

# Calculating the 95% confidence interval for the OR
lower_CI <- exp(beta_BMI - 1.96 * SE_BMI)
upper_CI <- exp(beta_BMI + 1.96 * SE_BMI)

# Difference in BMI for obese (at least 30 kg/m^2) relative to healthy-weight (22-24.99 kg/m^2)
bmi_difference_obese_to_healthy <- 6.505

# Adjusting the OR for obese vs. healthy-weight comparison
adjusted_OR_obese <- OR^bmi_difference_obese_to_healthy

# Adjusted confidence interval for the obese vs. healthy-weight comparison
adjusted_lower_CI_obese <- lower_CI^bmi_difference_obese_to_healthy
adjusted_upper_CI_obese <- upper_CI^bmi_difference_obese_to_healthy

# Printing the results
cat("Adjusted Odds Ratio for obese vs. healthy-weight males:", adjusted_OR_obese, "\n",
    "Adjusted 95% Confidence Interval:", adjusted_lower_CI_obese, "-", adjusted_upper_CI_obese, "
```

```
Adjusted Odds Ratio for obese vs. healthy-weight males: 3.760241
 Adjusted 95% Confidence Interval: 1.523728 - 9.279488
```

Interpreting this result, the model suggests a strong association between obesity and the increased odds of developing colorectal cancer, with age and smoking status held constant.

(e)

```r
# Residual deviance from the outputs
residual_deviance_model_1 <- 3675.3
residual_deviance_model_5 <- 3494.1

# Degrees of freedom for Model 1 and Model 5
# Model 1 has 2 parameters (intercept and BMI), Model 5 has 4 parameters (intercept, BMI, Age, an
df_model_1 <- 46  # This represents the residual degrees of freedom for Model 1
df_model_5 <- 43  # This represents the residual degrees of freedom for Model 5

# Calculate the test statistic D
D <- residual_deviance_model_1 - residual_deviance_model_5

# Degrees of freedom for the test
df <- df_model_1 - df_model_5

# Calculate the p-value
p_value <- pchisq(D, df, lower.tail = FALSE)

# Output the test statistic and p-value
cat("Likelihood Ratio Test Statistic (D):", D, "\n",
    "Degrees of Freedom for the Test:", df, "\n",
    "P-Value:", p_value, "\n")
```

```
Likelihood Ratio Test Statistic (D): 181.2
 Degrees of Freedom for the Test: 3
 P-Value: 4.856421e-39
```

The extremely small p-value indicates that we can reject the null hypothesis in favor of the alternative hypothesis.

Null Hypothesis (H_0): The simpler model (Model 1, which includes only BMI) is sufficient to explain the data.

Alternative Hypothesis (H_1): The full model (Model 5, which includes BMI, Age, and Smoking Status) provides a significantly better fit to the data.

This suggests that both Age and Smoking Status have a significant joint effect on the likelihood of colorectal cancer, after controlling for BMI.

(f)

```
# Coefficients from Model 5
intercept <- -6.44382
beta_BMI <- 0.20361
beta_Age <- 0.95990
beta_Smok <- 0.42122

# Values for the specific individual
BMI_value <- 23
Age_value <- 50
Smok_value <- 0  # 0 for non-smoker

# Calculate the linear predictor
linear_predictor <- intercept + (beta_BMI * BMI_value) + (beta_Age * Age_value) + (beta_Smok * Sm

# Calculate the probability using the logistic function
probability <- exp(linear_predictor) / (1 + exp(linear_predictor))

# Output the probability
probability
```

```
[1] 1
```